Investment Strategy HW5

Summary of Answers

- **Q1.** The risk-aversion formulation of mean-variance optimization is equivalent to the variance-capped expected-return maximization problem. For every strictly positive risk-aversion parameter, the optimizer is an efficient portfolio. The case `lambda = 0` corresponds to the maximum-return endpoint of the frontier, with the usual minimum-variance tie-break among return-maximizing portfolios. Conversely, every optimizer of the variance-constrained problem can be recovered from the risk-aversion problem by choosing the risk-aversion parameter equal to the KKT multiplier on the variance cap.
- **Q2(a).** Using the same two 3-year windows as in HW4 and excluding `GEHC` as instructed, the cross-sectional Pearson correlation between first-period and second-period annualized stock growth rates is only `0.04989`, and the Spearman rank correlation is `-0.00684`. Growth-rate persistence is therefore extremely weak and is even weaker than the `0.11068` mean-return stability correlation from HW4(k).
- **Q2(b).** The 50-stock daily rebalanced `1/n` portfolio over the second 3-year period produces cumulative return `74.99986%`, annualized return `19.78460%`, annualized volatility `14.16794%`, Sharpe ratio `1.39643`, maximum drawdown `-18.94706%`, CAGR `20.65684%`, average daily turnover `0.64266%`, and annualized turnover `161.94995%`.
- **Q2(c).** Relative to the HW4 long-only GMV portfolio, the rebalanced `1/n` portfolio earns higher cumulative return and a higher Sharpe ratio, but it also has higher volatility, a deeper maximum drawdown, a much higher beta to `SPY`, and materially higher trading intensity.

Supporting exhibits are provided as Appendix A through Appendix F:

- Appendix A: Q2(a) stock-level growth rates (`hw5_q2a_growth_rates.csv`)
- Appendix B: Q2(a) growth-rate summary (`hw5_q2a_growth_summary.csv`)
- Appendix C: Q2(b) equal-weight portfolio metrics (`hw5_q2b_equal_weight_metrics.csv`)
- Appendix D: Q2(c) equal-weight versus HW4 GMV comparison (`hw5_q2c_portfolio_comparison.csv`)
- Appendix E: compact summary metrics (`hw5_summary_metrics.csv`)
- Appendix F: selected code snippets from `code/hw5_analysis.py`

The compact cross-question metrics behind the summary bullets above are collected in Appendix E. The representative implementation snippets used in Sections 2(a) through 2(c) are collected in Appendix F.

Conventions and Data Choices

I use the same local adjusted-close Tiingo dataset and the same two subperiods as in HW4 so that the HW5 results are directly comparable to the prior assignment.

- First 3-year return window: `2020-04-14` through `2023-04-12` (`755` daily return observations)
- Second 3-year return window: `2023-04-13` through `2026-04-10` (`751` daily return observations)
- Annualization convention: `252` trading days
- Return convention: simple daily total returns from adjusted close
- Transaction costs: ignored unless otherwise stated

There is one important asset-treatment detail:

- In **Q2(a)**, the assignment explicitly allows me to ignore the stock with no data in the first 3-year period, so I exclude `GEHC` and work with the remaining `49` stocks.
- In **Q2(b)**, the assignment explicitly asks for a `1/n` portfolio with the `50` stocks. By the beginning of the second 3-year period, `GEHC` already has live trading history, so I include all `50` stocks there.
- In **Q2(c)**, I compare that `50` -stock equal-weight portfolio with the actual HW4 optimal Markowitz portfolio, which was constructed from the first 3-year estimation window and therefore necessarily excluded `GEHC` in estimation.

Data and Implementation Notes

- The main numerical calculations and CSV exhibits for Q2(a), Q2(b), and Q2(c) are implemented in `code/hw5_analysis.py` .
- The two SVG figures used in the write-up are generated by `code/generate_hw5_figures.py` .
- Appendix A through Appendix D correspond to the machine-generated exhibit files stored in `exhibits/` .
- Appendix E reproduces the compact summary-metrics exhibit inside the write-up for grading convenience.
- Appendix F reproduces representative code snippets from `code/hw5_analysis.py` so that the computational logic is visible in the write-up itself, as in HW4.

1. Different Formulations of the Mean-Variance Framework

I begin by rewriting the two optimization problems in compact notation.

Let the feasible set be

$$\mathcal{S} = \{x \in \mathbb{R}^n : \mathbf{1}'x = 1,\ x \geq 0\}.$$

The reduced-form expected-return maximization problem with a variance cap is

$$(\mathrm{P}_{\sigma_{\max}}) \qquad \max_{x \in \mathcal{S}} \ \mu' x \quad \text{subject to} \quad x' \Sigma x \leq \sigma_{\max}.$$

The risk-aversion formulation is

$$(\mathrm{U}_{\lambda}) \qquad \max_{x \in \mathcal{S}} \ \mu' x - \lambda x' \Sigma x, \qquad \lambda \in [0, \infty).$$

I now show carefully that these two formulations are equivalent descriptions of efficient portfolios.

Step 1. A Risk-Aversion Optimizer Solves a Variance-Constrained Return-Maximization Problem

Take any fixed risk-aversion parameter `lambda >= 0`, and let `x_lambda` be an optimizer of problem `U_lambda`.

Define the variance level actually chosen by that optimizer as

$$\bar{\sigma} = x_{\lambda}' \Sigma x_{\lambda}.$$

I claim that `x_lambda` must also solve the reduced-form return-maximization problem with variance cap equal to its own realized variance, that is,

$$\max_{x \in \mathcal{S}} \ \mu' x \quad \text{subject to} \quad x' \Sigma x \leq \bar{\sigma}.$$

To prove this, suppose the claim were false. Then there would exist some feasible portfolio `y` in the feasible set such that

$$y' \Sigma y \leq \bar{\sigma}$$

and

$$\mu' y > \mu' x_{\lambda}.$$

Now compare the objective values in problem `U_lambda`:

$$\mu' y - \lambda y' \Sigma y \geq \mu' y - \lambda \bar{\sigma}$$

because `y` satisfies the variance cap and `lambda >= 0`.

Since `y` has strictly higher expected return than `x_lambda`, we then obtain

$$\mu' y - \lambda y' \Sigma y > \mu' x_{\lambda} - \lambda \bar{\sigma}.$$

But by definition of the realized variance cap,

$$\mu' x_{\lambda} - \lambda \bar{\sigma} = \mu' x_{\lambda} - \lambda x_{\lambda}' \Sigma x_{\lambda}.$$

Therefore

$$\mu' y - \lambda y' \Sigma y > \mu' x_\lambda - \lambda x_\lambda' \Sigma x_\lambda,$$

which contradicts the fact that `x_lambda` is optimal for `U_lambda`.

So the contradiction proves that `x_lambda` indeed solves the reduced-form problem with variance cap equal to its own realized variance.

Step 2. Positive Risk Aversion Implies Efficiency; Zero Risk Aversion Gives the Frontier Endpoint

I first treat the case `lambda > 0` and show that `x_lambda` cannot be dominated by another feasible portfolio.

Suppose, for contradiction, that `x_lambda` were not efficient. Then there would exist some feasible portfolio `y` such that

$$\mu' y \geq \mu' x_\lambda,$$

$$y' \Sigma y \leq x_\lambda' \Sigma x_\lambda,$$

and at least one of these two inequalities would be strict.

If `lambda > 0`, then

$$\mu' y - \lambda y' \Sigma y \geq \mu' x_\lambda - \lambda x_\lambda' \Sigma x_\lambda,$$

and because at least one inequality is strict, the right-hand side is strictly improved:

$$\mu' y - \lambda y' \Sigma y > \mu' x_\lambda - \lambda x_\lambda' \Sigma x_\lambda.$$

That contradicts optimality of `x_lambda` for `U_lambda`.

This proves efficiency for every optimizer of `U_lambda` with `lambda > 0`.

If `lambda = 0`, then `U_0` becomes

$$\max_{x \in \mathcal{S}} \mu' x.$$

In that case, the objective only selects portfolios with maximal expected return. In general, if several portfolios attain the same maximal expected return, not every such maximizer must also have the smallest variance among that set. To match the efficient frontier exactly, I therefore interpret the `lambda = 0` case as follows: among all solutions to `U_0`, select one with the smallest variance.

Call that selected portfolio `x_0_min_sigma`. If there were some feasible portfolio `y` with

$$\mu' y \geq \mu' x_0^{\min \sigma},$$

$$y'\Sigma y \le (x_0^{\min\sigma})'\Sigma x_0^{\min\sigma},$$

and at least one strict inequality, then either:

- `y` has strictly higher expected return than `x_0_min_sigma`, contradicting the fact that `x_0_min_sigma` solves `U_0`; or
- `y` has the same expected return as `x_0_min_sigma` but strictly lower variance, contradicting the minimum-variance tie-break among return-maximizing portfolios.

So `x_0_min_sigma` is the maximum-return endpoint of the efficient frontier.

Step 3. Converse direction via the KKT conditions

The previous two steps show that every solution to `U_lambda` lies on the efficient frontier. To complete the equivalence, I now show that every solution to the variance-capped problem can be represented as a solution to `U_lambda` for an appropriate value of `lambda`.

Consider the variance-capped problem and write its Lagrangian:

$$\mathcal{L}(x,\gamma,\alpha,\nu) = \mu'x - \gamma(x'\Sigma x - \sigma_{\max}) + \alpha(\mathbf{1}'x - 1) + \nu'x,$$

where

- `gamma >= 0` is the multiplier on the variance constraint,
- `alpha` is the multiplier on the full-investment constraint,
- `nu >= 0` is the vector of multipliers on the long-only constraints.

The KKT first-order condition is

$$\mu - 2\gamma\Sigma x + \alpha\mathbf{1} + \nu = 0.$$

The remaining KKT conditions are

$$x'\Sigma x \le \sigma_{\max}, \qquad \mathbf{1}'x = 1, \qquad x \ge 0,$$
$$\gamma \ge 0, \qquad \nu \ge 0,$$
$$\gamma(x'\Sigma x - \sigma_{\max}) = 0,$$
$$\nu_i x_i = 0 \quad \text{for each } i.$$

Now look at `U_lambda`. Its Lagrangian is

$$\widetilde{\mathcal{L}}(x,\alpha,\nu) = \mu'x - \lambda x'\Sigma x + \alpha(\mathbf{1}'x - 1) + \nu'x,$$

and its first-order condition is

$$\mu - 2\lambda\Sigma x + \alpha\mathbf{1} + \nu = 0.$$

These conditions are the same as the KKT conditions of the variance-capped problem once I set

$$\lambda = \gamma^*,$$

where `gamma_star` is the optimal multiplier on the variance constraint.

There are two cases:

1. **The variance constraint binds.**
   Then `gamma_star > 0` is the shadow value of relaxing the variance cap, and choosing `lambda = gamma_star` makes the first-order conditions identical.
2. **The variance constraint does not bind.**
   Then complementary slackness implies `gamma_star = 0`, so the constrained problem reduces locally to pure expected-return maximization over the feasible set. That is exactly `U_0`; if there are multiple return-maximizers, the efficient-frontier endpoint is obtained by choosing the minimum-variance one among them.

Therefore every optimizer of the variance-capped problem is also an optimizer of `U_lambda` for an appropriate value of `lambda`.

Conclusion for Question 1

I have now shown both directions:

- Every optimizer of the risk-aversion problem `U_lambda` with `lambda > 0` solves a reduced-form return-maximization problem with the variance cap set equal to its own realized variance and is efficient.
- The case `lambda = 0` corresponds to the maximum-return endpoint of the frontier, with the usual minimum-variance tie-break among return-maximizing portfolios.
- Every optimizer of the reduced-form return-maximization problem with a variance cap solves the risk-aversion problem `U_lambda` for `lambda` equal to the KKT multiplier on the variance constraint.

So the two formulations are equivalent descriptions of the efficient frontier once the endpoint case `lambda = 0` is interpreted correctly. Put differently, the parameter `lambda` is not generating a different class of portfolios; it is simply another way to index efficient portfolios.

2. Portfolio Construction: Single Period vs. Rebalancing

2(a) Sharpe-Style Analysis for Stock Growth Rates

The assignment asks me to repeat the spirit of HW4(k), but now for stock growth rates rather than the daily mean returns used in HW4(k).

Because the note explicitly says I may ignore the stock with no first-period data, I exclude `GEHC` and work with the remaining `49` stocks.

For each stock index `i` and each subperiod `s in {1,2}`, let the daily adjusted-close total returns be denoted by `r_{i,t}^{(s)}` for `t = 1, ..., T_s`.

I define the realized gross growth over subperiod `s` as

$$\prod_{t=1}^{T_s}(1 + r_{i,t}^{(s)}).$$

I then annualize that realized growth rate using the same `252`-trading-day convention used in HW4:

$$g_i^{(s)} = \left[\prod_{t=1}^{T_s}(1 + r_{i,t}^{(s)})\right]^{252/T_s} - 1.$$

This gives one annualized realized growth rate for each stock in the first 3-year period and one annualized realized growth rate for the same stock in the second 3-year period.

Once these `49` pairs of growth rates are computed, I evaluate their cross-sectional stability exactly as in the Sharpe-style exercise: I compare the first-period growth vector `g_i^{(1)}` with the second-period growth vector `g_i^{(2)}`.

The Q2(a) growth-rate construction and stability calculations are implemented in `code/hw5_analysis.py`; Appendix F.1 reproduces the representative snippet. The code excerpt below is taken from that file:

```
growth_rows = []
first_growth = []
second_growth = []
for ticker in stocks_49:
    g1 = compounded_annual_growth([returns[ticker][date] for date in
first_dates])
    g2 = compounded_annual_growth([returns[ticker][date] for date in
second_dates_49])
    first_growth.append(g1)
    second_growth.append(g2)
    growth_rows.append({"Ticker": ticker, "FirstPeriodCAGR": g1,
"SecondPeriodCAGR": g2})

pearson = correlation(first_growth, second_growth)
spearman = correlation(ranks(first_growth), ranks(second_growth))
```

The same logic is reproduced again in Appendix F.1 together with its appendix heading.

Main numerical findings

The summary results are:

- number of stocks used: `49`
- first-period observations per stock: `755`
- second-period observations per stock: `751`
- Pearson correlation between first-period and second-period annualized growth rates: `0.04989`
- Spearman rank correlation: `-0.00684`
- same-sign count: `32` out of `49` stocks (`65.30612%`)
- average first-period annualized growth rate: `22.98247%`
- average second-period annualized growth rate: `17.56853%`
- median first-period annualized growth rate: `17.73356%`
- median second-period annualized growth rate: `14.39860%`
- average absolute change in annualized growth rate across periods: `26.25568` percentage points

For reference, the mean-return stability correlation from HW4(k) was `0.11068`. So the growth-rate persistence here is even weaker than the already-weak return persistence from HW4(k).

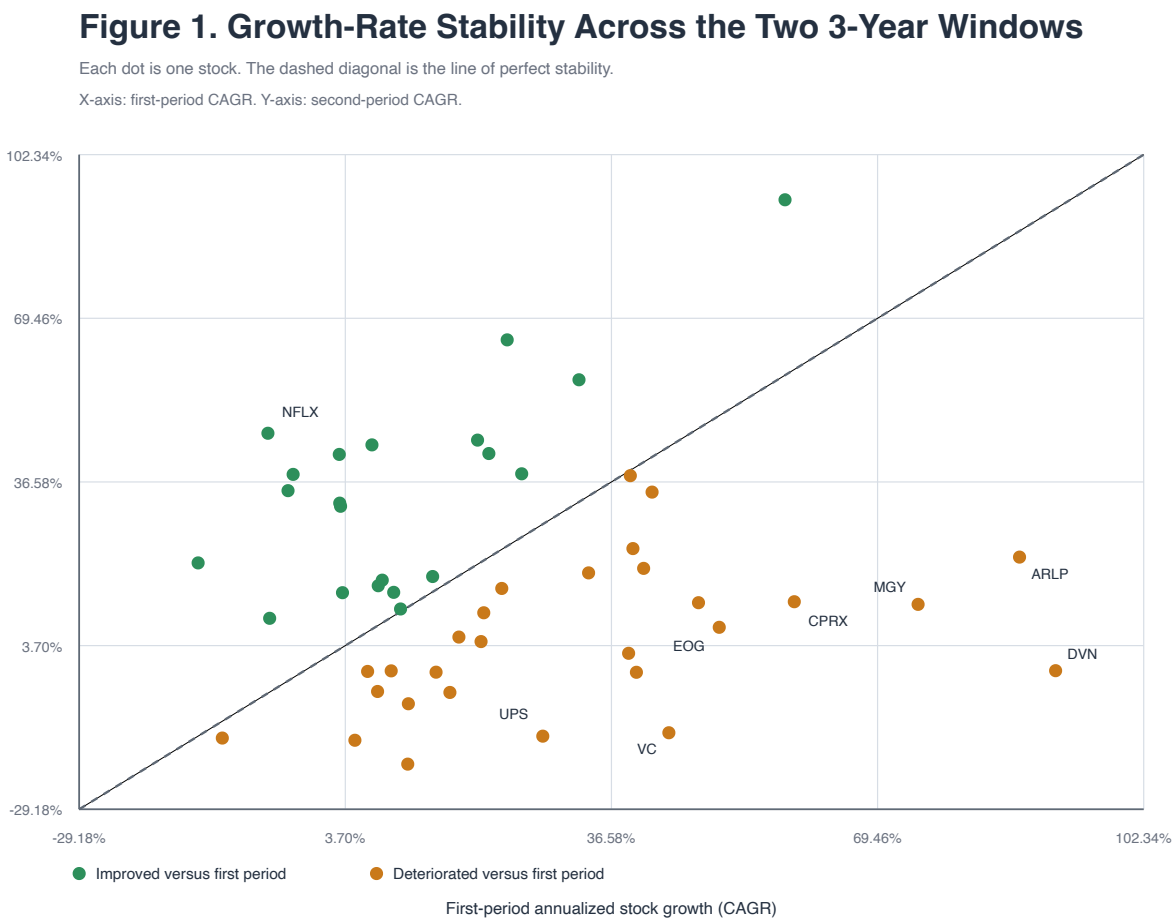
The compact metric summary reported in this subsection is provided as Appendix B (`hw5_q2a_growth_summary.csv`).

Representative stock examples

Table 1 gives a few representative reversals and improvements. The full stock-by-stock exhibit is provided as Appendix A (`hw5_q2a_growth_rates.csv`).

Ticker	First-Period CAGR	Second-Period CAGR	Change
DVN	91.45679%	-1.33910%	-92.79588 pp
ARLP	87.00016%	21.48065%	-65.51951 pp
VC	43.67419%	-13.81025%	-57.48444 pp
NFLX	-5.86314%	46.36306%	52.22620 pp
TSM	23.69819%	65.12551%	41.42732 pp
CGAU	2.95185%	42.11031%	39.15846 pp

Figure 1 plots first-period versus second-period annualized stock growth. If growth were stable across periods, the dots would cluster tightly around the diagonal. They clearly do not.



Does this make sense relative to HW4(k)?

Yes, it does make sense.

In HW4(k), my main conclusion was that expected-return estimates were much less stable across the two subperiods than volatilities and correlations. The current result is fully consistent with that conclusion, and in fact it is even stronger:

- the HW4(k) mean-return stability correlation was only `0.11068`;
- the HW5 growth-rate stability correlation is even lower at `0.04989`;
- the Spearman rank correlation is essentially zero (`−0.00684`), which means even the ranking of “growth winners” versus “growth losers” is not persistent.

Economically, this also makes sense. The first and second 3-year windows were very different market environments. Energy-related names such as `DVN`, `ARLP`, and `MGY` were exceptionally strong in the first window and much weaker in the second window. By contrast, several large technology and growth names such as `NFLX`, `TSM`, `META`, and `NVDA` were much stronger in the second window. So the weak persistence is not a numerical accident; it reflects genuine regime dependence in realized stock growth.

My conclusion for part (a) is therefore straightforward: realized stock growth over one 3-year period provides very little reliable information about realized stock growth over the next 3-year period in this sample. That conclusion is fully aligned with the warning from HW4(k) that return-related quantities are highly unstable across subperiods.

2(b) Daily Rebalanced `1/n` Portfolio in the Second 3-Year Period

For this part I follow the assignment literally and construct a `1/n` portfolio with the full `50` stocks at the beginning of the second 3-year period.

Since `n = 50`, the target weight on each stock is

$$w_i = \frac{1}{50} = 0.02000.$$

The portfolio is rebalanced daily, so after each trading day the weights are reset to `2.00000%` per stock for the following day.

Daily portfolio return formula

Let `r_{i,t}` denote the realized daily return of stock `i` on day `t` in the second 3-year period.

Because the beginning-of-day weights are equal each day, the portfolio return on day `t` is

$$r_{p,t}^{EW} = \sum_{i=1}^{50} w_i r_{i,t} = \frac{1}{50} \sum_{i=1}^{50} r_{i,t}.$$

The cumulative wealth process therefore evolves as

$$V_t = V_{t-1}(1 + r_{p,t}^{EW}),$$

with `V_0 = 1`.

Turnover formula

Daily rebalancing has a trading implication, so I also report turnover.

If the portfolio starts day `t` with equal weights `1/50`, then after the day's returns but before rebalancing, the drifted weight on stock `i` is

$$\widetilde{w}_{i,t} = \frac{\frac{1}{50}(1 + r_{i,t})}{1 + r_{p,t}^{EW}}.$$

To rebalance back to equal weights, the one-way turnover on day `t` is

$$\tau_t = \frac{1}{2} \sum_{i=1}^{50} \left| \widetilde{w}_{i,t} - \frac{1}{50} \right|.$$

I report both the average daily turnover and the corresponding annualized turnover, computed as `252` times the average daily turnover.

The daily rebalanced equal-weight portfolio and turnover calculations are implemented in `code/hw5_analysis.py`; Appendix F.2 reproduces the representative snippet. The code excerpt below is taken from that file:

```
r2_50 = [[returns[ticker][date] for ticker in all_stocks] for date in
second_dates_50]
eq_returns = [sum(row) / len(row) for row in r2_50]
eq_metrics = metrics(eq_returns)
eq_turnovers = [equal_weight_turnover(row) for row in r2_50]

def equal_weight_turnover(day_returns):
    n_assets = len(day_returns)
    target = 1.0 / n_assets
    post_weights_unnormalized = [target * (1.0 + ret) for ret in
day_returns]
    gross_return = sum(post_weights_unnormalized)
    post_weights = [weight / gross_return for weight in
post_weights_unnormalized]
    return 0.5 * sum(abs(weight - target) for weight in post_weights)
```

The same logic is reproduced again in Appendix F.2 together with its appendix heading.

Performance metrics

Using the `751` daily observations from `2023-04-13` through `2026-04-10`, the rebalanced equal-weight portfolio has:

- cumulative return: `74.99986%`
- annualized return: `19.78460%`
- annualized volatility: `14.16794%`
- Sharpe ratio: `1.39643`
- maximum drawdown: `-18.94706%`
- CAGR: `20.65684%`
- average daily turnover: `0.64266%`
- annualized turnover: `161.94995%`

I also compute two additional portfolio-characterization metrics that are relevant for part (c):

- beta to `SPY`: `0.84587`
- correlation with `SPY`: `0.90823`

Table 2 summarizes the equal-weight portfolio results.

Metric	Value
Observations	751
Cumulative Return	74.99986%
Annualized Return	19.78460%
Annualized Volatility	14.16794%
Sharpe Ratio	1.39643
Max Drawdown	-18.94706%
CAGR	20.65684%
Average Daily Turnover	0.64266%
Annualized Turnover	161.94995%
Beta to SPY	0.84587
Correlation to SPY	0.90823

The full CSV version of Table 2 is provided as Appendix C (`hw5_q2b_equal_weight_metrics.csv`).

These values indicate that the equal-weight portfolio delivered strong realized performance over this specific second-period window, but it did so with meaningful market exposure and a nontrivial rebalancing burden.

2(c) Comparison with the HW4 Optimal Markowitz Portfolio

The comparison portfolio from HW4 is the long-only global minimum-variance portfolio estimated from the first 3-year window and then held fixed through the second 3-year evaluation window.

From HW4, that portfolio's realized second-period performance was:

- cumulative return: `47.71862%`
- annualized return: `13.85000%`
- annualized volatility: `12.30208%`
- Sharpe ratio: `1.12583`
- maximum drawdown: `−12.38624%`
- CAGR: `13.98677%`

For a more informative portfolio-characterization comparison, I also report:

- beta to `SPY`: `0.33689`
- correlation with `SPY`: `0.41659`
- effective number of names: `6.94423`
- top-5 weight sum: `74.34459%`

By contrast, the equal-weight portfolio has:

- beta to `SPY`: `0.84587`
- correlation with `SPY`: `0.90823`
- effective number of names: exactly `50.00000`
- top-5 weight sum: exactly `10.00000%`

The Q2(c) comparison is assembled in `code/hw5_analysis.py`; Appendix F.3 reproduces the representative snippet. The code excerpt below is taken from that file:

```
gmv_weights = solve_gmv(sigma1_49)
gmv_returns = [
    sum(row[idx] * gmv_weights[idx] for idx in range(len(gmv_weights)))
    for row in r2_49
]

comparison_rows = [
    {
        "Portfolio": "Equal Weight 1/n",
        "Beta to SPY": covariance(eq_returns, spy_second) /
variance(spy_second),
        "Effective Names": 50.0,
        "Top-5 Weight Sum": 0.10000000,
    },
    {
        "Portfolio": "HW4 GMV",
        "Beta to SPY": covariance(gmv_returns, spy_second) /
variance(spy_second),
        "Effective Names": effective_names(gmv_weights),
        "Top-5 Weight Sum": sum(sorted(gmv_weights, reverse=True)[:5]),
    },
]
```

The same logic is reproduced again in Appendix F.3 together with its appendix heading.

Table 3 places the two portfolios side by side.

Metric	Equal-Weight `1/n`	HW4 GMV Portfolio
Assets Included	50	49
Rebalanced Daily	Yes	No
Cumulative Return	74.99986%	47.71862%
Annualized Return	19.78460%	13.85000%
Annualized Volatility	14.16794%	12.30208%
Sharpe Ratio	1.39643	1.12583
Max Drawdown	-18.94706%	-12.38624%
CAGR	20.65684%	13.98677%
Beta to SPY	0.84587	0.33689
Correlation to SPY	0.90823	0.41659
Effective Names	50.00000	6.94423
Top-5 Weight Sum	10.00000%	74.34459%
Average Daily Turnover	0.64266%	0.00000%
Annualized Turnover	161.94995%	0.00000%

The full CSV version of Table 3 is provided as Appendix D (`hw5_q2c_portfolio_comparison.csv`).

Figure 2 compares the equal-weight and HW4 GMV portfolios along the dimensions that matter most in part (c): performance, risk, market exposure, concentration, and trading intensity.

Direct numerical comparison

Relative to the HW4 GMV portfolio, the equal-weight portfolio has:

- `+27.28124` percentage points higher cumulative return
- `+5.93461` percentage points higher annualized return
- `+1.86586` percentage points higher annualized volatility
- `+0.27061` higher Sharpe ratio
- a drawdown deeper by `6.56083` percentage points
- `+6.67007` percentage points higher CAGR

Advantages of the rebalanced equal-weight portfolio

The equal-weight portfolio has several clear advantages in this realized second-period sample.

First, it performs better on the two headline reward metrics: realized return and realized Sharpe ratio. In this specific second-period window, a simple diversified `1/n` rule beats the optimized

GMV rule on both total performance and risk-adjusted performance.

Second, it is much less exposed to estimation error. The equal-weight portfolio does not require estimating a covariance matrix and then solving an optimizer that can amplify small estimation differences. That robustness matters because HW4(k) already showed that return-related quantities were not stable across periods.

Third, it is far more diversified in weight concentration. The equal-weight portfolio spreads capital uniformly across `50` stocks, whereas the GMV portfolio is highly concentrated. Its effective number of names is only `6.94423`, and nearly three-quarters of the capital (`74.34459%`) sit in the top five positions. That concentration makes the GMV portfolio more vulnerable to model error and to stock-specific shocks in a small number of names.

Fourth, this specific second-period sample rewarded broad equity exposure. The equal-weight portfolio's beta to `SPY` is `0.84587`, so it participated much more fully in the realized market upside than the GMV portfolio, whose beta was only `0.33689`.

Advantages of the HW4 GMV portfolio

The GMV portfolio also has important advantages.

First, it is meaningfully more defensive. Its beta to `SPY` is only `0.33689`, versus `0.84587` for the equal-weight portfolio. Its correlation with `SPY` is also much lower (`0.41659` versus `0.90823`). So the GMV portfolio behaves much less like a broad equity market portfolio.

Second, it achieves lower realized volatility and a much smaller drawdown. Even though it underperforms on return, it fulfills its intended role as a low-variance portfolio: realized volatility is only `12.30208%`, versus `14.16794%` for equal weight, and the maximum drawdown is materially smaller in magnitude (`−12.38624%` versus `−18.94706%`).

Third, it requires essentially no rebalancing during the second-period holding window in the way I implemented HW4. That means its realized return is not exposed to the same level of turnover-related transaction-cost risk as the equal-weight portfolio, whose annualized turnover is `161.94995%`.

Interpretation

The realized second-period evidence suggests the following.

If the investor cares most about realized return and realized Sharpe ratio over this particular evaluation window, the daily rebalanced equal-weight portfolio looks better. It is simple, broadly diversified, and it outperforms the GMV portfolio in this sample.

If the investor instead cares more about downside protection, low market exposure, and lower implementation trading, then the HW4 GMV portfolio still has a defensible advantage. Its lower

volatility, lower drawdown, and much lower beta are not accidents; they are exactly what a minimum-variance allocation is designed to deliver.

So my bottom-line comparison is:

- the equal-weight portfolio is the stronger performer in this realized second-period sample;
- the GMV portfolio is the more defensive and lower-implementation-risk portfolio;
- the superiority of equal weight here should not be interpreted as a general theorem, because this comparison is period-specific and strongly affected by the realized market environment.

Appendices

Appendix A through Appendix F are reproduced below inside the write-up so that the PDF version shows both the supporting data tables and the representative implementation logic in one document. The original exhibit files remain available in `exhibits/`.

Appendix A. Q2(a) Stock-Level Growth Rates (`hw5_q2a_growth_rates.csv`)

Table A1 reproduces the full stock-level growth-rate exhibit used in Q2(a).

Ticker	FirstPeriodCAGR	SecondPeriodCAGR	Difference	SameSign
AAPL	0.33755159	0.18307777	-0.15447382	Yes
ADBE	0.04883661	-0.15319378	-0.20203040	No
ALLE	0.03350296	0.14331771	0.10981475	Yes
AMGN	0.07765721	0.15729733	0.07964012	Yes
AMZN	-0.03379393	0.34831384	0.38210778	No
ANF	0.32569736	0.57109083	0.24539347	Yes
ARLP	0.87000164	0.21480654	-0.65519510	Yes
BMY	0.09366411	-0.01376898	-0.10743309	No
CAH	0.21433968	0.42289648	0.20855680	Yes
CALM	0.14488367	0.17579673	0.03091307	Yes
CBT	0.38700957	0.02153102	-0.36547856	Yes
CF	0.40556194	0.19215279	-0.21340915	Yes
CGAU	0.02951847	0.42110305	0.39158459	Yes
CPRX	0.59170981	0.12515359	-0.46655622	Yes
DECK	0.47334969	0.12319496	-0.35015474	Yes

Ticker	FirstPeriodCAGR	SecondPeriodCAGR	Difference	SameSign
DVN	0.91456786	-0.01339099	-0.92795885	No
EOG	0.49898767	0.07375612	-0.42523154	Yes
EXEL	0.03112417	0.31691157	0.28578740	Yes
GNTX	0.07684605	-0.05517498	-0.13202104	No
GOOG	0.20037670	0.44985577	0.24947907	Yes
HCA	0.39233538	0.23193071	-0.16040467	Yes
HRB	0.39664839	-0.01663481	-0.41328320	No
ICLR	0.11412898	-0.20113201	-0.31526100	No
INCY	-0.05648442	0.09183027	0.14831469	No
JNJ	0.08261849	0.16847907	0.08586058	Yes
LLY	0.38912918	0.37841869	-0.01071049	Yes
MCK	0.41599163	0.34538851	-0.07060312	Yes
META	0.06988690	0.44027442	0.37038752	Yes
MGY	0.74465975	0.11983470	-0.62482505	Yes
MRK	0.17733556	0.05411756	-0.12321800	Yes
MSFT	0.20804951	0.10295063	-0.10509889	Yes
NEM	-0.02748811	0.38097520	0.40846332	No
NFLX	-0.05863140	0.46363056	0.52226195	No
NVDA	0.58025794	0.93272078	0.35246284	Yes
ORCL	0.23029182	0.15181890	-0.07847292	Yes
PEGA	-0.14487251	0.20312888	0.34800139	No
PEP	0.14910197	-0.01642419	-0.16552616	No
PFE	0.11489092	-0.07984822	-0.19473914	No
PYPL	-0.11498431	-0.14875982	-0.03377551	Yes
QCOM	0.20468606	0.04506346	-0.15962261	Yes
RNR	0.09676932	0.14398603	0.04721671	Yes
SYF	0.25491181	0.38216870	0.12725688	Yes
TROW	0.06453240	-0.01490406	-0.07943645	No
TSM	0.23698187	0.65125507	0.41427320	Yes
UBER	0.03009171	0.32322366	0.29313196	Yes
UHS	0.10524891	0.11056409	0.00531518	Yes

Ticker	FirstPeriodCAGR	SecondPeriodCAGR	Difference	SameSign
UPS	0.28097253	-0.14495933	-0.42593186	No
VC	0.43674187	-0.13810253	-0.57484439	No
YELP	0.16621748	-0.05714423	-0.22336171	No

Appendix B. Q2(a) Growth-Rate Summary (`hw5_q2a_growth_summary.csv`)

Table B1 reproduces the compact summary exhibit for Q2(a).

Metric	Value
Stock Count	49.00000000
First-Period Observations	755.00000000
Second-Period Observations	751.00000000
Growth Pearson Correlation	0.04989366
Growth Spearman Correlation	-0.00683673
Same-Sign Count	32.00000000
Average First-Period CAGR	0.22982474
Average Second-Period CAGR	0.17568527
Median First-Period CAGR	0.17733556
Median Second-Period CAGR	0.14398603
Average Absolute CAGR Change	0.26255675
HW4 Mean-Return Stability Corr	0.11067852

Appendix C. Q2(b) Equal-Weight Portfolio Metrics (`hw5_q2b_equal_weight_metrics.csv`)

Table C1 reproduces the full equal-weight portfolio metrics exhibit for Q2(b).

Observations	Cumulative Return	Annualized Return	Annualized Volatility	Sharpe Ratio	Max Drawdown	CA
751	0.74999860	0.19784604	0.14167941	1.39643468	-0.18947061	0.2

Appendix D. Q2(c) Portfolio Comparison (`hw5_q2c_portfolio_comparison.csv`)

Table D1 reproduces the full portfolio-comparison exhibit for Q2(c).

Portfolio	AssetsIncluded	RebalancedDaily	Cumulative Return	Annualized Return	Annualized Volatility
Equal Weight 1/n	50	Yes	0.74999860	0.19784604	0.14167941
HW4 GMV	49	No	0.47718621	0.13849989	0.12302078

Appendix E. Compact Summary Metrics

Table E1 reproduces the compact summary-metrics exhibit in the body of the write-up.

Item	Value
Q2(a) growth Pearson correlation	0.04989366
Q2(a) growth Spearman correlation	-0.00683673
Q2(a) same-sign count	32
Q2(b) cumulative return	74.99986%
Q2(b) annualized return	19.78460%
Q2(b) annualized volatility	14.16794%
Q2(b) Sharpe ratio	1.39643468
Q2(b) max drawdown	-18.94706%
Q2(c) equal-weight minus GMV annualized return	5.934615 percentage points
Q2(c) equal-weight minus GMV Sharpe ratio	0.27060951

Appendix F. Selected Code Snippets from `code/hw5_analysis.py`

This appendix collects the main implementation snippets used to compute the stock-growth stability statistics, the daily rebalanced equal-weight portfolio metrics, and the HW4 GMV comparison. I include them here so that the write-up follows the same basic style as HW4: the

main text remains readable, but the core computational logic is still visible inside the document itself.

Appendix F.1 Q2(a) Growth-Rate Construction and Stability

```
growth_rows = []
first_growth = []
second_growth = []
for ticker in stocks_49:
    g1 = compounded_annual_growth([returns[ticker][date] for date in
first_dates])
    g2 = compounded_annual_growth([returns[ticker][date] for date in
second_dates_49])
    first_growth.append(g1)
    second_growth.append(g2)
    growth_rows.append({"Ticker": ticker, "FirstPeriodCAGR": g1,
"SecondPeriodCAGR": g2})

pearson = correlation(first_growth, second_growth)
spearman = correlation(ranks(first_growth), ranks(second_growth))
```

Appendix F.2 Q2(b) Equal-Weight Returns and Turnover

```
r2_50 = [[returns[ticker][date] for ticker in all_stocks] for date in
second_dates_50]
eq_returns = [sum(row) / len(row) for row in r2_50]
eq_metrics = metrics(eq_returns)
eq_turnovers = [equal_weight_turnover(row) for row in r2_50]

def equal_weight_turnover(day_returns):
    n_assets = len(day_returns)
    target = 1.0 / n_assets
    post_weights_unnormalized = [target * (1.0 + ret) for ret in
day_returns]
    gross_return = sum(post_weights_unnormalized)
    post_weights = [weight / gross_return for weight in
post_weights_unnormalized]
    return 0.5 * sum(abs(weight - target) for weight in post_weights)
```

Appendix F.3 Q2(c) Equal-Weight versus GMV Comparison

```
gmv_weights = solve_gmv(sigma1_49)
gmv_returns = [
    sum(row[idx] * gmv_weights[idx] for idx in range(len(gmv_weights)))
```

```
    for row in r2_49
]

comparison_rows = [
    {
        "Portfolio": "Equal Weight 1/n",
        "Beta to SPY": covariance(eq_returns, spy_second) /
variance(spy_second),
        "Effective Names": 50.0,
        "Top-5 Weight Sum": 0.10000000,
    },
    {
        "Portfolio": "HW4 GMV",
        "Beta to SPY": covariance(gmv_returns, spy_second) /
variance(spy_second),
        "Effective Names": effective_names(gmv_weights),
        "Top-5 Weight Sum": sum(sorted(gmv_weights, reverse=True)[:5]),
    },
]
```